DATATHON EXECUTIVE SUMMARY

Predictive Triage Engine for Nigerian PHCs

41,778	25,237	2,056	774
RAW FACILITIES INGESTED	PUBLIC PHCS MONITORED	CRITICAL PRIORITY FACILITIES	LGAS RANKED NATIONALLY

1. THE SPECIFIC PROBLEM & WHY IT MATTERS

Primary Healthcare Centres (PHCs) are the first point of care for millions of people across Nigeria. However, health administrators often face sudden stockouts of basic medicine and vaccines. This happens because they lack real-time information about facility operations. Many health posts fail to report inventory levels regularly, creating dangerous **operational blind spots**. Without an early warning system, government health teams only react after shortages occur. Our project solves this problem by building a data-driven triage engine that identifies high-risk health centres before supplies run out.

2. APPROACH & DATA METHODOLOGY

We used the national **GRID3 NGA Health Facility v3.0 dataset**. We cleaned the data and filtered it down from 41,778 total facilities to an isolated cohort of **25,237 public PHCs**. To measure vulnerability, we created a composite **Shortage & Blind Spot Risk Score (0–100)** using three weighted pillars:

1. Operational Blind Spots (40%)
Measures unknown functional status and missing reporting logs.

2. Supply Chain Instability (40%)
Tracks historical stockout frequencies over a 6-month period.

3. Geographic Isolation (20%)
Calculates distance and travel friction to central supply stores.

Finally, we aggregated facility risk scores to rank all **774 Local Government Areas (LGAs)** nationwide, allowing decision-makers to prioritize supply delivery where it is needed most.

3. WHAT IS WORKING VS. WHAT IS SIMULATED

Fully Functional Features	Simulated / Placeholder Components
Data Ingestion Pipeline: Cleaned and processed 25,237 public PHCs from the raw GRID3 registry.	**Reporting Log Latency:** Synthesized field report delays using realistic statistical distributions.
LGA Triage Engine: Ranked all 774 LGAs by intervention urgency, spotting key hotspots in Jigawa, Gombe, Kaduna, Borno, and Delta.	**Stockout Frequencies:** Simulated 6-month medicine stockout counts prior to live API database links.
Interactive Dashboard: Developed a functional Streamlit web application with state filters, risk maps, and downloadable CSV reports.	**Spatial Travel Friction:** Applied Euclidean distance metrics instead of real-time road network travel times.

4. WHAT WE WOULD BUILD NEXT

- **Live API Integrations:** Connect the risk engine directly to DHIS2 and eLMIS software to receive real-time supply logs and automated stock alerts.
- **Predictive Machine Learning:** Train XGBoost and LightGBM time-series models to forecast stockout risks 30 days in advance.
- **Road Network Friction Mapping:** Incorporate seasonal road quality and weather data to calculate actual delivery travel times.
- **Automated Reorder Alerts:** Send instant SMS notifications to regional logistics managers when an LGA enters the critical risk zone.